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Juvenile Friends, Behavior, and Immune Responses to Separation in Bonnet Macaque Infants

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BOCCIA, M. L., J. M. SCANLAN, M. L. LAUDENSLAGER, C. L. BERGER, A. HIJAZI AND M. L. REITE. *Juvenile friends, behavior, and immune responses to separation in bonnet macaque infants.* PHYSIOL BEHAV **61**(2) 191–198, 1997.—Individual differences in the response to maternal separation in nonhuman primate infants have been attributed to (among other variables) presence or absence of processes that may model social support in humans. Alternative attachments to other members of the social group buffer the infant against a depressive response to maternal separation. This hypothesis was tested in a group of bonnet macaques by manipulating the presence or absence of alternative juvenile attachment figures (friends) during separation. Infants who retained such attachments showed fewer behavioral evidences of depression when separated from their mothers. These infants without friends also showed changes in lymphocyte activation by mitogens or natural cytotoxicity that were not evident in the infants with juvenile friends. Across all separated infants, natural cytotoxicity was positively correlated with juvenile affiliative behavior directed toward the infants during the separation. These results support the hypothesis that social support, available from alternative attachments, can modulate the response to loss, and can account for some of the individual differences seen in these responses. *Copyright © 1997 Elsevier Science Inc.*

Bonnet monkey Psychoneuroimmunology Social support Natural cytotoxicity Mitogen response Animal models

THE role of social support in buffering humans against adverse health consequences, both physical and mental, of loss has received considerable attention by researchers (e.g., 8,10,14, 20,37). These studies have found that people with greater social support have lower morbidity and mortality risk. The exact relationship between health and support, however, and particularly the presumed mediating role of the immune system, is not as well studied. An animal model of this system, which would require a species having complex social systems similar to humans, would facilitate investigation of this relationship.

In young nonhuman primates, both early (nonsocial) experiences and the social environment are important in shaping the development of social bonds and attachments, as well as infant responses to social separation experiences. For example, the presence of social support, in the form of alternative attachments, may mitigate the psychobiological reaction to maternal loss (4). This study addresses the hypothesis that the presence of alternative social attachments during separation affect the infant's biobehavioral response to loss.

Comparative studies of attachment and separation responses have revealed both similarities and differences among different species of primates. The classic protest-despair response was initially identified in nonhuman primates in rhesus macaques (33). Similar responses to separation have been identified in pigtail macaques (17,30), chimpanzees (1,34), orangutans (22), and gorillas (23). Neither bonnet macaques (17,19,27,31) nor langurs (11) typically exhibit the despair phase.

The origin of these species differences has been attributed to a variety of mechanisms, including differences in the availability of other, prior-existing social attachments (18). We, with others (e.g., 13,17,38), view attachment as a particular kind of relationship involving an emotional bond between individuals and being evident in behaviors that serve to establish and maintain contact or proximity, and marked by evidence of distress and/or depression when the individuals are separated from each other. Although the primary attachment is typically formed with the mother, alternative attachments, as evidenced by similar behaviors, may develop between other individuals and the infant, if opportunities for this are available.

When separated from their mothers, the infants of these species presumably utilize these alternative attachments for social support, thereby attenuating the depressive response (11,17,19,

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TABLE 1
BEHAVIORAL TAXONOMY

Behavior Category	Behavior	Description
Maternal	Away	Beyond arms length of the mother
	Cradle	On the ventrum of the mother (not facing away) with one, both, or neither of the mother's arms around the infant
	Touch	In physical contact with mother, but not Cradle
	Proximity	Within arms length of the mother, and making no physical contact
	Wean	Removal or denial of access to the nipple
Other social contact	No contact	Beyond arms length of any other animal in the group
	Ventral other	Cradle by animal other than the mother
	Touch other	Physical contact with animal other than the mother
	Proximity other	Proximity to animal other than the mother
	Play	Playful interaction with another animal, including chasing, wrestling, etc.
	Groom	Oral, manual or visual inspection of another animal's hair, skin, teeth, etc.
Environmentally directed behavior	Object exploration	Manual or oral manipulation, or close visual inspection, of any object in the pen, not play
	Eat/drink	Any ingestion of food or water
	Self play	Play not involving other animals, including motor play (e.g., swinging from perch) or object play (playful, repetitive manipulation of objects)
	Locomotion	At least two steps in any direction
Other nonsocial behaviors	Sleep/drowse	Sleeping or quiet rest
	Self orality	Oral contact with any body part, not grooming
	Slouch	Depressive posture, involving lethargy, collapsed posture and depressive facial expression
	Stereotypy	Repetitive, idiosyncratic behaviors (e.g., pacing, head flips)
Agonistic behaviors	Contact aggression	Bite, hit or otherwise attempt to harm another animal, with physical contact
	Threat	Open mouthed facial expression signalling aggressive intent
	Submit	Flee, crouch, cringe, and/or fear grimace

31). In both bonnet macaques and langurs, mothers are less restrictive with their infants than the other, previously mentioned, species and the infants interact with and develop relationships with other group members (in laboratory studies of other species, siblings and aunts are often absent, and these relationships are less likely to develop). In both species, when the mother is removed, the infants actively seek out and maintain contact with others in the social group. Allomaternal behavior or adoption in the absence of this prior attachment is ineffective in preventing the loss-induced depressive response (26,29).

We have previously demonstrated that manipulation of the early social environment in a bonnet macaque social group may result in infants developing more restrictive social networks than typical in this species (4). Although limited to 2 subjects, this study was highly suggestive of the role of social support. When separated from its mother, the infant with a more restricted social network exhibited a greater depressive response than the infant with a broader social network. This suggested that the presence of alternative attachments could buffer infants against the biobehavioral effects of loss.

In the present study, we examined the effects of the presence of social support during a separation experience on behavioral and immunological responses to separation in bonnet macaque infants. To accomplish this, we allowed species-typical social attachments to develop in a group of young macaques, and then randomly selected half the infants to retain a juvenile friend; all other conspecifics were removed from the group.

METHODS

Subjects

Subjects were 6 bonnet macaque infants born in a social group composed of 1 adult male, 6 adult females, the infants, and 6 juveniles, each being an older sibling of one of the subject infants. The infants were 4 to 6 months old and the juveniles were 18 to 24 months old. The group was housed in a $2.1 \times 2.5 \times 4.0$ m pen with glazed cinderblock walls, wire mesh ceilings, and fiberglass shelves and plastic pipes in place for sitting and climbing. A one-way mirror permitted unobtrusive observations. Water was always available, and feeding occurred at 0900 h daily, with supplemental fruit provided on weekdays.

Procedures

Blood sample collection. Three ml of heparinized blood (10 μ/ml blood) was drawn from the saphenous vein of unanesthetized subjects. Each infant was removed from its social group, and placed under light physical restraint in an individual restraint cage. Blood samples were obtained for the entire group of infants in less than 30 min. Six baseline samples were collected prior to the separation at approximately 4-day intervals to ensure habituation to the sampling procedure. Only the last 3 baseline samples were used in analysis. Samples were also collected 2 h and 48 h after separation, and at weekly intervals for 2 weeks.

Behavioral data collection. For 2 weeks immediately prior to the separation, observations of the frequency of social interactions were collected, using 5-min focal animal sampling (39) of a number of social behaviors, including play, contact, proximity, and grooming. Both the behavior and the social partner were recorded for a Round Robin Analysis of Variance, described below. These samples were collected 5 days per week, 4 times per day (twice in the morning, and twice in the afternoon). Prior to the start of observations, interobserver reliability was established at a minimum of 85% for each category of behavior. For most categories of behavior, reliability exceeded 90%.

During the baseline and separation, additional behavioral observations were conducted, collecting 5-min focal animal sam-

ples of both duration and frequency of social and nonsocial behaviors (see taxonomy in Table 1). These observations were made twice daily (once in the morning and once in the afternoon), except on the first day of separation when 6 observation sessions were conducted (3 in the morning and 3 in the afternoon). After collection of baseline data, the infants were separated for 2 weeks from their social group. Three juveniles, identified by social network analysis (see below) who had significant, complex relationships with 3 randomly selected infants (age matched to nonsupport infants) were left with the separated infants.

All behavioral data were collected directly on-line, in real time, using an IBM-compatible computer system called The Observer (24), which permits the collection of sequence, frequency, and duration of all behaviors listed in Table 1. The behavior taxonomy was based originally on that of Kaufman and Rosenblum (16) and modified by Reite and Short (28), and most recently by Laudenslager et al. (19).

Separation procedure. The separation occurred when the infants were 4 to 6 months old. Obsevations included 2 weeks of baseline and 2 weeks of separation.

The separation was accomplished at 0800 h on the Monday of the first week of separation. The infants was separated by removing the entire group from the home pen, and separating each infant from its mother. This was accomplished by placing the mother under light mechanical restraint and allowing the infant to run into another small transport cage. The infant was then returned to the home pen. The remainder of the social group was housed in a similar pen at the opposite end of the laboratory out of visual, auditory, and olfactory contact with the separated infants and juveniles.

Data Analysis

Assessment of preexistent relationships. To identify with which individuals the infants established relationships, the social network data were subjected to Round Robin Analyses of Variance (4,9,35), which examines the social preference network of animals living in a social group.

This analytic scheme permits investigation of a number of dimensions of social relationships as described by Hinde (13), such as content, reciprocity, and complexity. These dimensions reflect the degree and quality of the attachment, and were used to identify juveniles with significant complex relationships with 1 of the 3 infants chosen to retain social support during the separation.

The significant social preferences of all 6 infants for the juveniles are presented in Table 2. The infants chosen to have support during separation where selected randomly, but matched for age with the infants separated without the presence of social support.

Assessing response to separation. Focal animal behavioral observations were analyzed using a 2-way, mixed design ANOVA, with experimental condition (support, no support) as a between-subjects variable and phase of separation (baseline, separation day 1 [SD1], separation) as a within-subjects variable. Immune data were also subjected to ANOVAs with support as a between-subjects variable and phase of separation (2 h, 48 h, and 1 week) as within-subjects variable. In addition, Pearson Product Moment Correlations were computed between total number of affiliative behaviors directed by the juveniles during the separation to each infant, and B- and T-cell proliferative responses, natural cytotoxicity, differential cell counts, and total WBC. All tests were 2-tailed tests of significance.

Immunological Assays

Mononuclear cell isolation. Three ml of heparinized blood was obtained, as described above. Samples were spun at 600 *g*, after 100 μl was removed for a differential cell count and a total white cell count determined on a Coulter counter (Model ZBI). Plasma was removed and stored at −20°C. Cells were resuspended in Dulbecco's phosphate buffered saline (DPBS) and overlaid on 3 ml of Ficoll Paque (Pharmacia) for the isolation of mononuclear cells as previously described [see (19) for details]. Recovered mononuclear cells were washed 3 times in DPBS and adjusted to either 6.0×10^6/ml (cytotoxicity assay) or 1.0×10^6/ml (lymphocyte stimulation) in supplemented (L-glutamine, penicillin/streptomycin, fetal calf serum) RPMI-1640 with HEPES media. For the lymphocyte stimulation assay, 5% fetal calf serum (FCS) was used, whereas the cytotoxicity assay media included 10% FCS and $2X10^5$M 2-mercaptoethanol.

Lymphocyte stimulation. Mononuclear cells were stimulated in triplicate in the wells of a flat bottomed microtiter plate at 1×10^5 cells/well with the mitogens phytohemaggluttin (PHA) (0.1, 0.25, and 0.5 μg/well), concanavalin A (ConA) (0.5, 0.75, 1.0 μg/well), and pokeweed (PWM) (1.0, 2.0, and 3.0 μg/well) in a total volume of 200 μl. Following incubation at 37°C, 5% CO_2 in a humidified CO_2 incubator for 72 hr, cells were pulsed with 1.0 μCi ^{3}H-thymidine in 10 μl saline and harvested 4 h later onto glass-fiber filter discs with a Skatron cell harvester. Filter discs were counted on a liquid scintillation counter in 2.0 ml of Optifluor™ (Packard). Results were calculated as:

(CPM experimental stimulated culture) − (CPM background + CPM media control unstimulated culture).

The median of the 3 triplicate determinations was used in the statistical analysis.

TABLE 2

SOCIAL PREFERENCES FOR ALL 6 INFANTS OF THE JUVENILES REMAINING BEHIND WITH THE INFANTS DURING THE SEPARATION (I, Initiate; R, receive)

	Infant No.						
Juvenile	3	5	25 Support	35	53 Support	72 Support	Support
2		Rec; Agon	RCont; RProx				Present
7					RProx; IPlay; RPlay	IProx; RProx; IPlay; RPlay	Present
24						IProx	
37	RProx; RPlay; RAgon	RCont; RProx	RProx			RProx	
51		ICont; RProx	RCont; RProx	RProx	RCont; RPlay	RPlay	
76						IAgon; IProx; RProx	Present

Natural cytotoxicity. Target cells were obtained from K562 human myelogenous leukemia cell line maintained in supplemented RPMI-1640 at 37°C in a humidified CO_2 (5%) incubator. Target cells were either fed or split as necessary 24–48 h prior to labelling with ^{51}Cr. Labeling consisted of centrifuging approximately 1×10^7 cells from the culture flask, removing the supernatant, and resuspending the cells in 100 μl FCS to which 100 μl of 100 μCi ^{51}Cr in unsupplemented media was added. Targets were incubated for 1 h at 37°C in a waterbath. Targets were centrifuged at 150 *g* for 5 min at 5°C and resuspended in cold Hank's balanced salt solution (HBSS) and incubated for 30 min on ice. Targets were then washed and centrifuged 3 times with cold HBSS at 150 *g*, 15 min, 0°C for each wash. Viability of targets was determined with Eosin Red and cell concentration was adjusted to 1×10^5 targets/ml of supplemented media. Effectors were prepared as indicated above. Effector-to-target ratios of 60:1, 30:1, 15:1, and 7.5:1 were plated in triplicate into the wells of a round bottomed microtiter plate to a total volume of 200 μl. Plates were centrifuged at 35 *g* for 5 min at room temperature before incubation. Suspensions were then incubated for 4 h at 37°C in a humidified CO_2 (5%) incubator. At the conclusion of the incubation period, plates were centrifuged at 70 *g* for 10 min at room temperature. To 2 ml of Optifluor™, 100 μl of the supernatant from each well was added. Control cultures (spontaneous release) consisted of 1×10^4 targets without effectors in 200 μl media. Total CPM was determined after lysing 1×10^4 targets with ml ZAPoglobulin™ (Coulter). The percent lysis at each effector-to-target ratio was determined from the following formula:

$$\text{Percent lysis} = \frac{\text{CPM experimental} - \text{CPM spontaneous}}{\text{Total CPM} - \text{CPM spontaneous}} \times 100$$

RESULTS

All the infants showed behavioral changes in response to the separation, but the infants without support were affected to a greater extent. In addition, there were immunological changes following separation evident in the unsupported infants that were not evident in the supported infants. These included changes in B- and T-cell proliferative responses and natural cytotoxicity. Finally, changes in some immune measures (most notably cytotoxicity) were directly related to the amount of affiliative behaviors the separated infants received from the juveniles during the separation.

Behavior

All the infants showed an increase in Ventral Contact over the separation period (SD 1 = 4 s/5 min interval, SW 2 = 26; $F(3/12) = 31.29$, $p = 0.000$), and a concomitant decrease in other kinds of Contact (SD 1 = 68 s/5-min interval, SW 2 = 21; $F(3/12) = 7.62$, $p = 0.004$) and Proximity (SD 1 = 106 s/5-min interval, SW 2 = 84; $F(3/12) = 21.03$, $p = 0.000$), suggesting increasing allomaternal behavior on the part of the juveniles as the separation progressed.

All infants also exhibited decreased Object Exploration ($F(3/12) = 3.93$, $p = 0.036$), Locomotion ($F(3/12) = 16/09$, $p = 0.000$), Sleep ($F(3/12) = 30.41$, $p = 0.000$), and Social Play ($F(3/12) = 8.30$, $p = 0.003$), and increased Eat ($F(3/12) = 20.24$, $p = 0.000$) (see Table 3). These changes in behavior are typical of the depressive phase of infant monkeys' response to separation.

Infants with support, however, did not show an increase in Passive behaviors, unlike the unsupported infants ($F(3/12) = 3.11$, $p = 0.06$). Infants with support also did not show the decrease in Self Play exhibited by the unsupported infants ($F(3/12) = 3.22$, $p = 0.06$) (see Fig. 1). Thus, there were indications that the presence of support ameliorated the depressive response. Infants with support also appeared to behave more aggressively toward the infants without social support during the separation, as indicated by higher initiate Threat ($F(1/4) = 7.33$, $p = 0.05$) and initiate total Aggression ($F(1/4) = 10.70$, $p = 0.03$), and lower initiate Submission ($F(1/4) = 8.64$, $p = 0.04$) in these infants. It is not clear why this was the case, although it suggests that the infants without support had the added stress of being the objects of this aggression.

TABLE 3

BEHAVIORAL CHANGES DURING SEPARATION THAT AFFECTED ALL INFANTS, REGARDLESS OF THE PRESENCE OR ABSENCE OF SUPPORT

Behavior	Baseline	SD1	SW1	SW2
Object exploration	51	51	25	21
Eat	29	16	68	25
Locomote	126	108	66	38
Sleep	15	0	41	131
Social play	57.4	13.4	0.9	1.7

Data are presented as s per 300-s observation period.

Immunology

Following separation, there were significant differences between supported and unsupported infants on a number of the immune measures. Supported infants showed greater proliferative responses to ConA and PWM, whereas unsupported infants showed less proliferative responses to these mitogens ($F(12/48) = 2.65$, $p = 0.008$, and $F(12/48) = 2.80$, $p = 0.06$, respectively) (see Fig. 2). Infants without support also exhibited an initial decrease in natural cytotoxicity 2 h following separation not evident in infants with support ($F(9/36) = 1.78$, $p < 0.10$; see Fig. 3). Unsupported infants' natural cytotoxicity gradually returned to baseline by the end of 1 week of separation.

We also examined the relationship between the amount of support provided by the juveniles (operationalized as the total number of affiliative behaviors directed toward the infants) and the immune measures (see Tables 4 and 5). Natural cytotoxicity was positively correlated with the amount of support provided throughout the separation. Thus, high cytotoxicity was associated with greater levels of social support. Although the results were less dramatic for the lymphocyte response to mitogen stimulation, there were, nevertheless, significant positive correlations between affiliative behavior and the magnitude of this response, particularly for ConA.

DISCUSSION

The present observations lend credence to the notion that the presence of social support mitigates against the adverse mental and physical health consequences of social stressors, particularly loss (14). Perhaps more importantly, they suggest a possible mechanism for increased morbidity and mortality associated with the lack of social support, particularly related to infectious illness or cancer, that may reside in changes in immunoregulation. The lack of social support in patients with cardiovascular disease is a risk factor for poor prognosis following myocardial infarct (10), which is independent of other medically prognostic factors

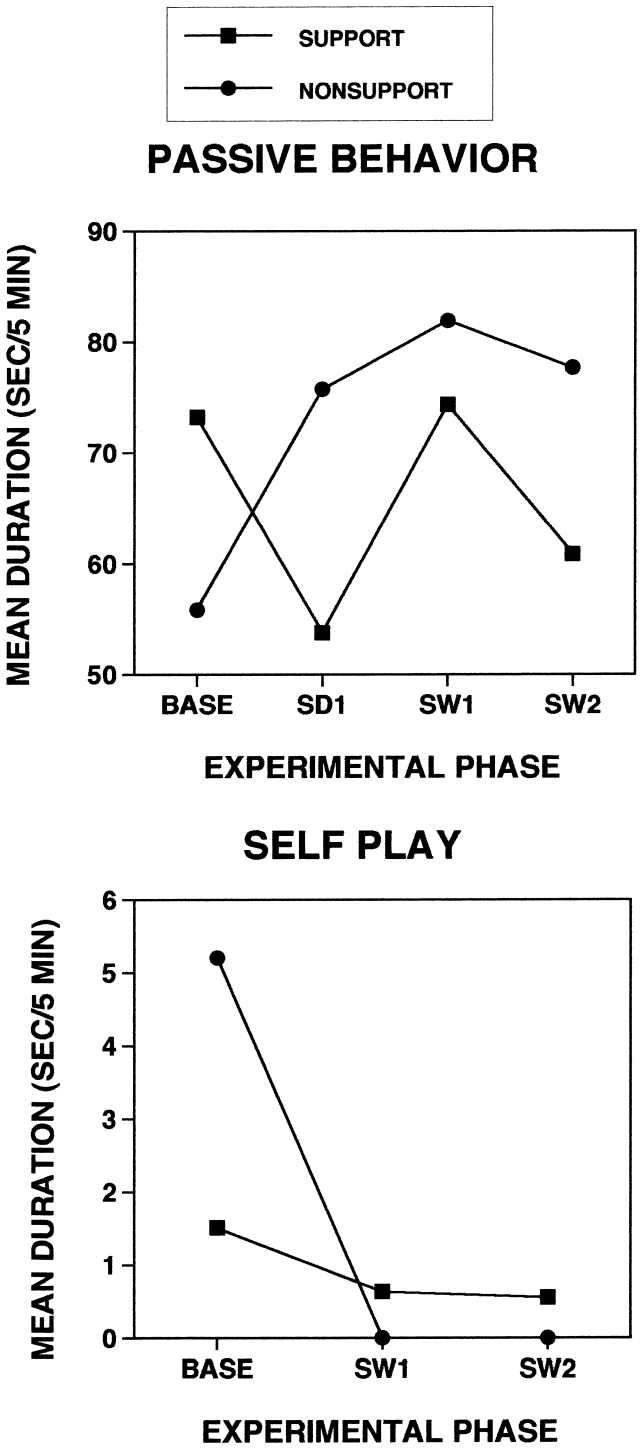


FIG. 1. Levels of Passive and Self Play as a function of the presence or absence of social support. BASE, baseline; SD1, first day of separation; SW1, remaining days of the first week of separation; SW2, second week of separation.

(37). Of direct relevance to the present study, spouses of cancer patients having the greatest level of social support were observed to have higher levels of natural cytotoxicity (2). In healthy first- and second-year medical students, levels of natural cytotoxicity

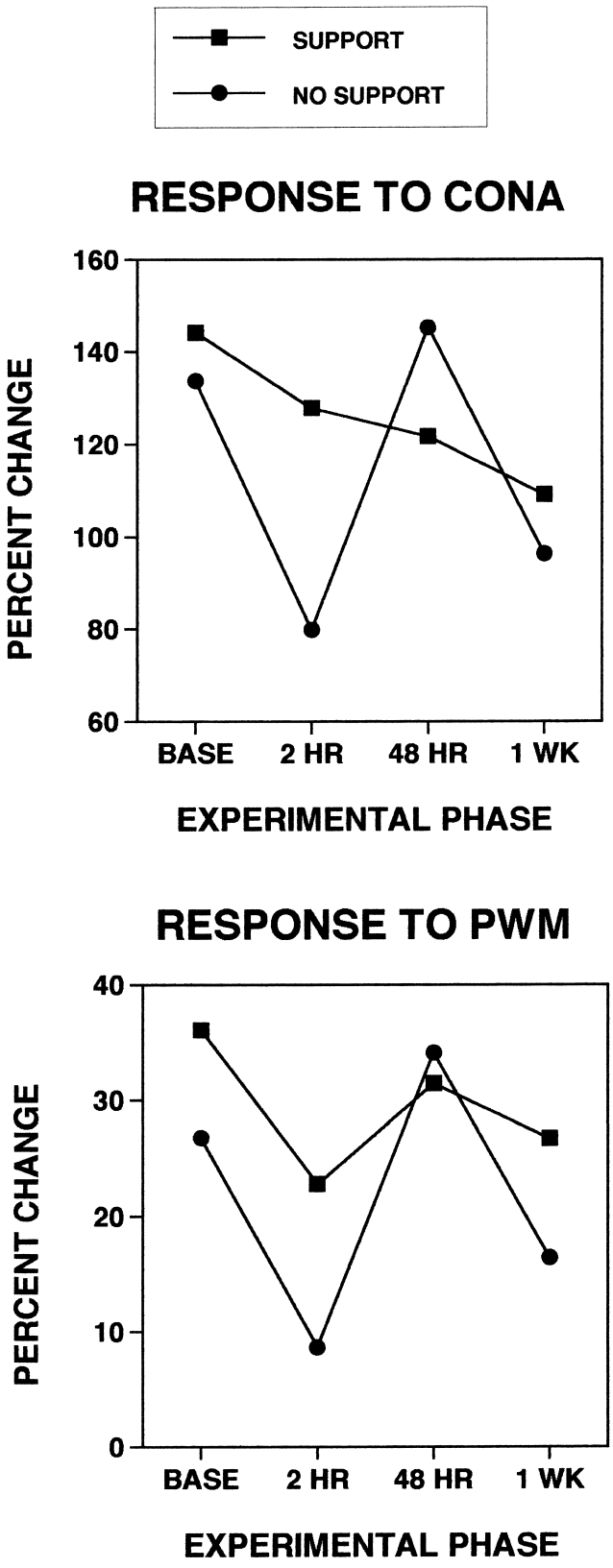


FIG. 2. Lymphocyte proliferation in response to ConA (upper graph) and PWM (lower graph), over the course of the separation.

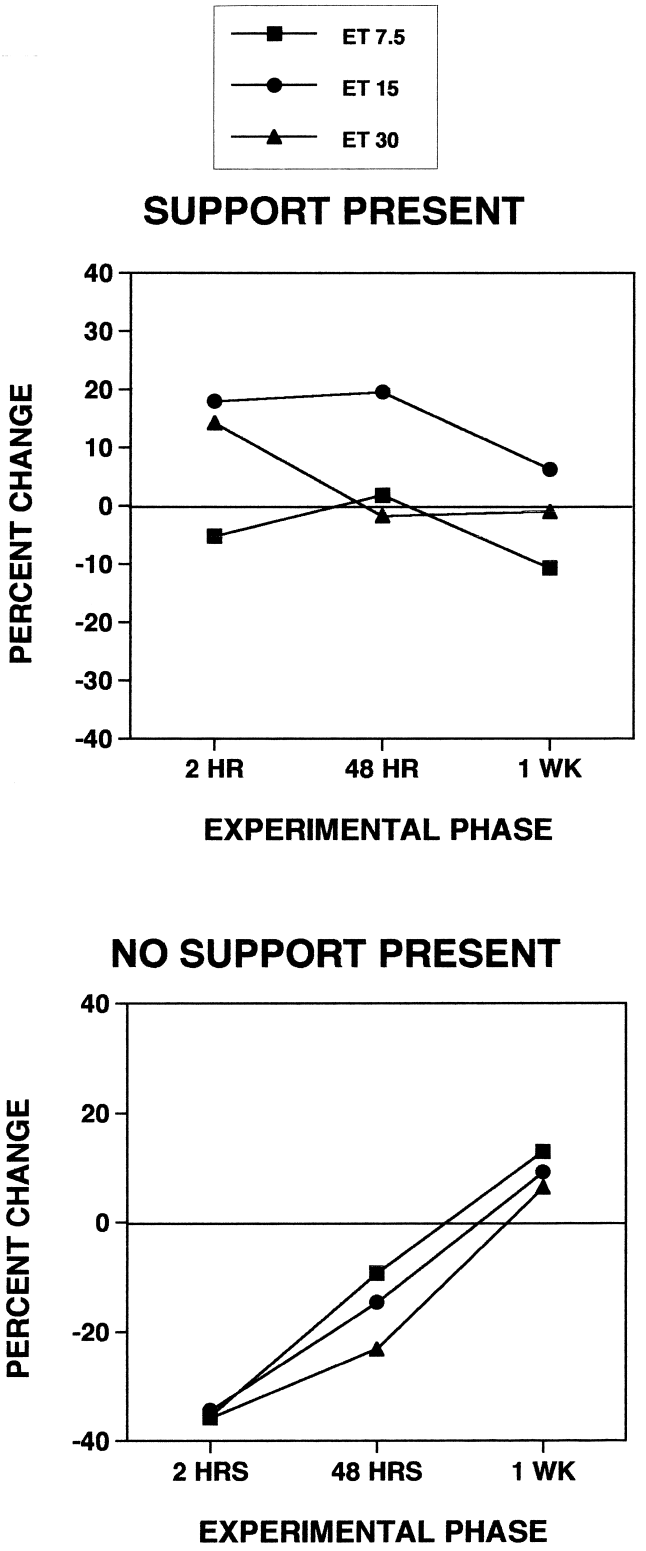


FIG. 3. Change in natural cytotoxicity over the course of the separation in infants with (upper graph) and without (lower graph) social support.

were directly related to self-reported social support (32). When the seeking of social support as a coping style and the perceived quality of support are entered in to a multivariate model to predict NK activity in breast cancer patients, these factors contribute significantly (20) but, not surprisingly, they do not account for all of the variance. The application of animal models of social support, in which healthy subjects are studied in well-controlled environments may shed light on its impact on immunoregulation.

Species differences in response to maternal separation have been attributed by others to differences in allomaternal behavior and the availability of social support (11,17): In bonnet macaque groups, mothers are typically less restrictive with their infants, and others engage in a great deal of allomaternal and play behavior and, thus, develop attachments with them. The social support provided by these alternative attachments thereby buffers the bonnet infant against maternal loss-induced depression. In the bonnet macaques studied here, infants with no social support showed greater behavioral depression, and greater changes in immune measures than infants with social support. Furthermore, the amount of support provided by the juveniles (operationalized as the amount of affiliative behavior directed toward the infants during the separation) was significantly correlated with both lymphocyte proliferative responses and natural cytotoxicity.

The present observations of significant relationships between behavior and the immune response, although from relatively few subjects, concur with previous observations from our laboratory (5,6). It was noted following immunization with a foreign protein that plasma levels of specific antibody (IgM class) to that protein were positively correlated with affiliative behavior received by infant macaques (both bonnet and pigtail) following immunization. Affiliative behaviors included both ventral contact and proximity to others. Thus, infants receiving greater social support showed higher levels of IgM class antibodies to the foreign protein. The IgG response was inversely related to behaviors reflecting behavioral distress. These studies suggest that social support represented by affiliative behaviors directed toward an infant may blunt the immunological consequences of adverse psychosocial experiences.

TABLE 4

CORRELATIONS BETWEEN B- AND T-CELL PROLIFERATIVE RESPONSES AND DIFFERENTIAL CELL COUNTS, AND JUVENILE AFFILIATIVE BEHAVIORS (CONTACT, CRADLE, PLAY, GROOM, PROXIMITY) DIRECTED TOWARD THE INFANTS DURING SEPARATION

Mitogen/Concentration	2 h	48 h	1 Week
PHA			
1	+0.79*	+0.97†	+0.85†
2.5	–	+0.88†	–
5.0	–	–	–
ConA			
1	–	+0.75*	
5	+0.79*	+0.92*	+0.81†
7.5	+0.76*	–	+0.73*
10	+0.78*	–	–
PWM			
1	–	–	–
5	–	–	–
10	–	+0.72*	–
20	+0.83†	+0.75†	–

* $p < 0.10$; † $p < 0.05$; ‡ $p < 0.01$.

TABLE 5

CORRELATION BETWEEN NATURAL CYTOTOXICITY AND JUVENILE AFFILIATIVE BEHAVIORS (AS IN TABLE 4) DIRECTED TOWARD THE INFANTS DURING SEPARATION

Effector/Target Ratio	2 h After Separation	48 h After Separation	1 Week After Separation
7.5:1	+0.84†	+0.95‡	+0.82†
15:1	+0.87†	+0.98‡	+0.80*
30:1	+0.84†	+0.93‡	+0.80†
60:1	+0.83†	+0.87†	+0.79*

* $p < 0.10$; † $p < 0.05$; ‡ $p < 0.01$.

The time-course of the present changes in immune measures differs from a previous report from this laboratory (19). When pigtail infants were separated from their mothers by removing the mother from the social group, changes in lymphocyte activation were not noted until the second week of separation. Little or no change was noted for the bonnet infants, which was attributed to the presence of previously established alternative attachments during the separation from the mother. These alternative attachments were not present for the pigtail monkey infants. In the present study, we noted changes by 2 h postseparation for the bonnet infants lacking a juvenile friend. These changes returned to baseline by the end of the first week of separation. Thus, compared to our data from pigtail infants, these bonnet infants showed faster changes (by 2 h) that returned to baseline faster (by the end of the first week of separation). These differences may be explained by differences in the type of separation: Removing all adults from the group was much more disruptive than removing 1 female. The rapid return to baseline levels may be due to the increased affiliative behavior directed to all infants by the end of the first week of separation.

The biological origins of the differences in immunological responses to loss seen between infants with and infants without social support is an important question. Several possible candidates exist that have been demonstrated both to be affected by maternal separation and to modulate immune-system function. These include aspects of glucocorticoid, catecholamine, and endogenous opiate regulation. Plasma cortisol has been demonstrated to increase significantly during maternal separation in a number of primate species (3,36) and has also been shown to affect immune regulation (21). Catecholamines and related compounds, such as neuropeptide Y, that also modulate immune function (12,40), are affected by maternal separation, both at the adrenal medullary level (7) and in the cerebrospinal fluid (CSF) (3). Finally, endogenous opiates, both affect immune function (25) and have been implicated in separation response (15). The links between these physiological parameters, however, and social support and behavioral responses to loss have not been demonstrated.

In summary, the present study suggests that the presence of social support during separation can modulate both behavioral and immunological responses to loss. Furthermore, the amount of social support (operationalized as the amount of affiliative behavior directed toward the infants) correlates with the degree of immune modulation, specifically with respect to natural cytotoxicity and lymphocyte proliferation. The mechanisms linking these phenomena have not yet been elucidated.

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